

## Commentary

# Improving safety in radiation oncology

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As a radiation oncologist, I am proud of the importance our specialty has historically placed on quality assurance and patient safety. Accordingly, I view the events of the last year as a reminder of its importance to our field and an opportunity for us to update our approach to these issues. Dr Marks and his co-authors should be commended for producing such a comprehensive and thoughtful review on the subject.<sup>1</sup> In this commentary, I would like to highlight some of the issues that I feel are particularly worthy of future action.

The article appropriately starts with a comprehensive review of existing data on error rates in radiation oncology. I found the relative paucity of these data and the nonuniform manner in which they have been collected striking. Contrast these data with the number of reports on the treatment of prostate cancer with radiation therapy using the consensus definition of prostate specific antigen failure published in a single month! If we are going to have a clear understanding of the magnitude of the problem and where we need to intervene (and if the intervention was successful), then we need better error and near-miss reporting data collection and reporting systems at both the national and departmental levels. I agree wholeheartedly with the authors' suggestion that such systems will allow for greater awareness across the specialty of serious, (but sometimes very rare) events, and for local improvements, respectively. It will also greatly facilitate safety-related research in radiation oncology.

In reading the next section of the article detailing both the traditional and new challenges that can lead to errors in radiation oncology, I was struck not only by their large number, but also by how they are so broad and multifaceted. Given the overwhelming nature of the list, it was reassuring to see the long list of potential interventions offered by the authors to address these concerns. The challenge going forward seems to be to identify which interventions are going to be most effective in radiation oncology, and then to pursue them aggressively. The hierarchy of effectiveness conceptual framework, which was mentioned several times by the authors, may help us to prioritize our efforts. Learning from Lean, we need to keep those activities that add value to current practices and let go of those that do not. Looking at the hierarchy, it is clear that some of the improvements will need to come from the device and software manufacturers whose products we rely on to treat our patients. However, we should not sit back and wait for them to act, but instead we should be working amongst ourselves to identify "low-tech" safety best practices.

Given our past, I am optimistic that significant improvements will continue to be made in how safely we treat our patients. However, because I am a realist, I am also aware that a great deal of work is needed to be done to achieve this goal. For the sake of our patients, let us all get to work!

## Reference

1. Marks LB, Jackson M, Xie L, et al. The challenge of maximizing safety in radiation oncology. *Practical Radiat Oncol.* 2011;1(1): 2-14.

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